MODULE 5: Deep Fakes

Objective
To help participants understand:
- What deepfakes are.
- How they are created using modern technology at a conceptual level.
- Why they can be more dangerous than other forms of fabricated content.
- The importance of recognising deepfakes and remaining vigilant as citizens.

5.1 What Is a Deepfake?
A deepfake is a fabricated or manipulated video, audio recording, or image created using artificial intelligence (AI) and machine learning that can appear or sound highly realistic.
A defining characteristic of deepfakes is that they can be difficult to distinguish from genuine content through sight or hearing alone.

Examples:
- Replacing one person's face with another's in a video or image.
- Imitating someone's voice to make them appear to say something they never said.
- Presenting a statement or event that never occurred as real.

5.2 How Are Deepfakes Created? A Conceptual Overview
This section provides a simple explanation without exploring technical complexities.
Creating a deepfake generally involves:
- Collecting images, videos, or audio recordings of a person.
- Using AI to learn patterns in that person's facial movements, voice, and expressions.
- Generating new video or audio that appears authentic.

Key point:
Deepfake creation relies on AI-based tools and techniques, which distinguish it from conventional editing.

5.3 Why Are Deepfakes So Convincing?
Deepfakes can appear convincing for several reasons:
- Facial expressions and voices can be imitated with remarkable accuracy.
- Lighting, shadows, and expressions can appear realistic.
- People often place greater trust in video and audio than in written information.
- People tend to think, "I saw it with my own eyes and heard it with my own ears."

As a result, people may accept what they see and hear without questioning its authenticity.

5.4 Why Are Deepfakes Dangerous?
Deepfakes do more than spread false information; they can undermine the foundations of trust.

At the Individual Level
- Reputational damage.
- Psychological stress.
- Risks to personal safety.

At the Social Level
- Confusion.
- Distrust.
- Rumours and panic.

At the Democratic and Civic Level
- Voters' decisions may be influenced.
- Trust in elections and civic processes may decline.
- Misconceptions may contribute to political instability and social tensions.

5.5 Deepfake Risks in the Bangladesh Context
The risks posed by deepfakes are growing in Bangladesh because of:
- The popularity of video content on social media.
- The tendency to share content quickly.
- Limited public understanding of the technology.

Deepfakes can create serious confusion, particularly when they concern elections, public figures, social reputation, or sensitive issues.

5.6 Early Warning Signs of Deepfakes
Although deepfakes can be difficult to identify, potential warning signs include:
- Unnatural facial expressions.
- Lip movements that do not match the speech.
- Unusual blinking patterns.
- A voice that sounds monotonous or artificial.
- An unclear source for the video or audio.

Remember: Not all deepfakes display these signs. Any doubts about authenticity should prompt further verification.

5.7 Deepfakes and the Crisis of Trust
One of the most serious harms caused by deepfakes is that people may lose confidence in their ability to distinguish genuine content from fabricated material.
As a result:
- Genuine events may also be viewed with suspicion.
- People may develop the belief that "nothing can be trusted."
- Democratic debate and civic dialogue may be weakened.

Do:
- Assess video or audio critically before accepting it as authentic.
- Verify the source.
- Prioritise information from official and reliable sources.
- Refrain from sharing when in doubt.

Don't:
- Rush to share sensational videos.
- Assume a claim is true simply because "there is video evidence."
- Make emotionally charged comments.
- Express opinions before verifying the information.